Question: What are the tradeoffs between verification and invasiveness for different interventions?"

Purpose

Who is this for and why?
A policy expert/org, government/regulator, oversight body, safety organisations etc that may have strong ideas about pacing and coordinating pacing between frontier labs and tech giants, but perhaps lack ideas on what's possible on the verification/enforcement/monitoring side of things. As well as context as to the invasiveness of various verification methods and the likely pushback, hesitancy and even incentives that may cause frontier actors to circumvent such measures.

Scope

Worth thinking about what topics are in-scope and what are out-of-scope. Open to rethinking this.

In-scope

- Needs some anchoring to popular (substantiated by well known orgs or even frontier labs themselves) or well-thought out hypothetical pacing policies.
- Introductory background (could just be a paragraph or spread throughout) to broad stages of development of a new AI model: pre-training, post-training, safety evaluations, deployment (staged, comprehensive) etc.
- What can be paced specifically, this can mean training compute caps, compute growth cap, capability caps (stop scaling when capability hits a certain benchmark), dangerous capability limits (stopping when dangerous capabilities are shown), full training pauses, RSI ban (limiting use of AI to perform AI R&D)
- What are methods/techniques that can be used to verify or even enforce pacing. Use examples of proposed methods (compute verification) and existing oversight methods that may draw from other domains like AI safety. Add context on existing frontier lab-third party relationships that share data, such as AI safety organisations and government agencies. Pacing proposals often stop before ideating a full verification/enforcement/monitoring mechanism, so it's worth being creative here.
- The level of invasiveness of these methods and techniques and the likelihood of buy-in by the labs.
- The risk/drawbacks of verification methods and their invasiveness: risk of data leaks, espionage/corporate espionage, ways for labs to "legally" or to the letter of the policy, circumvent the restrictions but violating the spirit of it, ways for labs to outright cheat and

avoid detection. Possible perverse incentives caused by pacing verification/monitoring/enforcement methods: causing labs to do some other thing that essentially furthers AI capabilities anyway.
- Worth having some recommendation of feasible/tractable verification techniques (which will be tied to specific pacing policies, even though I would consider it out-of-scope to recommending specific pacing policies too strongly, that would be other articles/outputs). Could present them in a tiered fashion, levels of invasiveness/pushback.

Out-of-scope

- Comprehensive description of all or even most pacing policies that have been proposed
- Determining the merits of specific pacing policies
- Discussing the eagerness/hesitancy of frontier labs to pace, this paper should have a starting point of assuming labs are open to some pacing regime generally
- Answering the "China" question more broadly, i.e. "what if China doesn't pace". **It can be worth as a bonus to include how international pacing agreement might work, though I suspect that should also be a different output, even though verification is very much the hard question even if agreement in principle is possible.**
- Answering "What if ABC lab doesn't agree to pacing", goes back to my 3rd point, this would be the responsibility of another output. This paper assumes all frontier labs are open to some pacing, or that we can implement some form of intervention that doesn't even require buy-in from the labs (including new/future labs, tech giants that may take pacing as an opportunity to catch up).
- Discussing pacing ideas that focus on parties other than the frontier labs: cloud/neoclouds/datacenters, energy providers, data providers, GPU/hardware manufacturers. **(Open to changing my mind on this).** Verification mechanisms could most certainly be applied to the products/services of these third parties, but its context must be grounded in their interactions with the frontier labs.

Output ideas

Output should be a guide, but that itself encompasses many possible outputs. Can be a well organised essay/article type guide. A cheat sheet neatly structured. A lengthy whitepaper. A clickable web experience, like a wiki, with expandable context/information/extra details. Coordination with the other Pacing-tech groups would be helpful.